CUSTOMER ENGAGEMENT AND PRODUCT UTILIZATION ANALYTICS FOR RETENTION STRATEGY

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BACKGROUND AND CONTEXT

Banks increasingly recognize that customer behaviour and engagement – not just demographics- long-term retention. Customers may appear financially strong (high balance or salary) but still churn due to:

Low Engagement

Low engagement in a customer engagement and product utilization analytics is for the customers who do not want to buy or need the product because of low performance of bank employees that's a disconnection between tracking raw performance indicators and delivering behavioural insights for retention.

The causes of low engagement are siloed data that tracks usage metrics like logins or page views separately from customer relationship data hiding true churn risks.

Over dependence on Demographics focuses on long-term retention strategy depends upon age, income, high balance, salary, size and structure of human population.

Limited Product Adoption

Limited product adoption indicates high churn risk that affects the number of employees attrition whose performance level is low as compared to customers who do have a high balance ratio.

Weak relationship depth with the bank

Customers who do have a high balance ratio has an in-depth knowledge of financial products that may appear financially strong which is a low churn risk. Employees that are customers who relies upon a single product or low digital interaction has a weak relationship in depth with the bank.

Understanding how customers use banking products and services is essential to designing: -

Cross-sell strategies: -

Customers who want to buy a banking product or a service such as personal banking gets benefits in addition to a relatable item like a financial product such as mutual funds, investment plans and withdrawal plans which is known as a cross-sell strategy.

Loyalty programs: -

Loyalty programs like reward programs, gifts, perks or ceremonies that banks do initiate to maintain a healthy relationship with the customers which reduces attrition rate and retain customers at any cost.

A Relationship Strength Index (RSI) which composites a metric combines activity levels, product ownership and feature adoption to measure true loyalty.

- RSI = Engagement score + Product score + Credit card score

- **Engagement-driven retention strategies**

Engagement driven retention strategies drive emotional attachments and motivation to customers for long term loyalty and reducing attrition rate to make them feel prioritized customers.
Retention campaigns aim to increase customer lifetime value (CLV) and lower the overall churn rate that are designed to keep prior customers encouraged, engaged and growing in value over time.

PROBLEM STATEMENT

Despite having data on customer engagement and product usage, banks often lack:

- **Quantitative insight into which behaviours drive retention**

Quantitative analysis quantifies a specific action, usage frequency and interaction pattern that differentiates with long term customer retention or user retention. It reveals an active user takes an action to use a product or a service frequently shows a pattern to interact with end active users.

Retention rate: $\frac{\text{Active users at end} - \text{New users during period}}{\text{Active users at start}} * 100$

- **Clarity on whether product depth reduces churn**

The product in depth knowledge about the feature adoption, variation, description, awareness and mobility that shows a clarity on how a product is well known for that helps the customers being tracked out of composite metrics which reduces the attrition rate.
The difference between the product variation and product feature adoption is the depth which reduces the churn rate.

- **Evidence on whether high balances alone ensure loyalty**

Customers who maintain higher account balances are more likely to switch the banks instead of being in a competitive advantage with the lower account balance users.

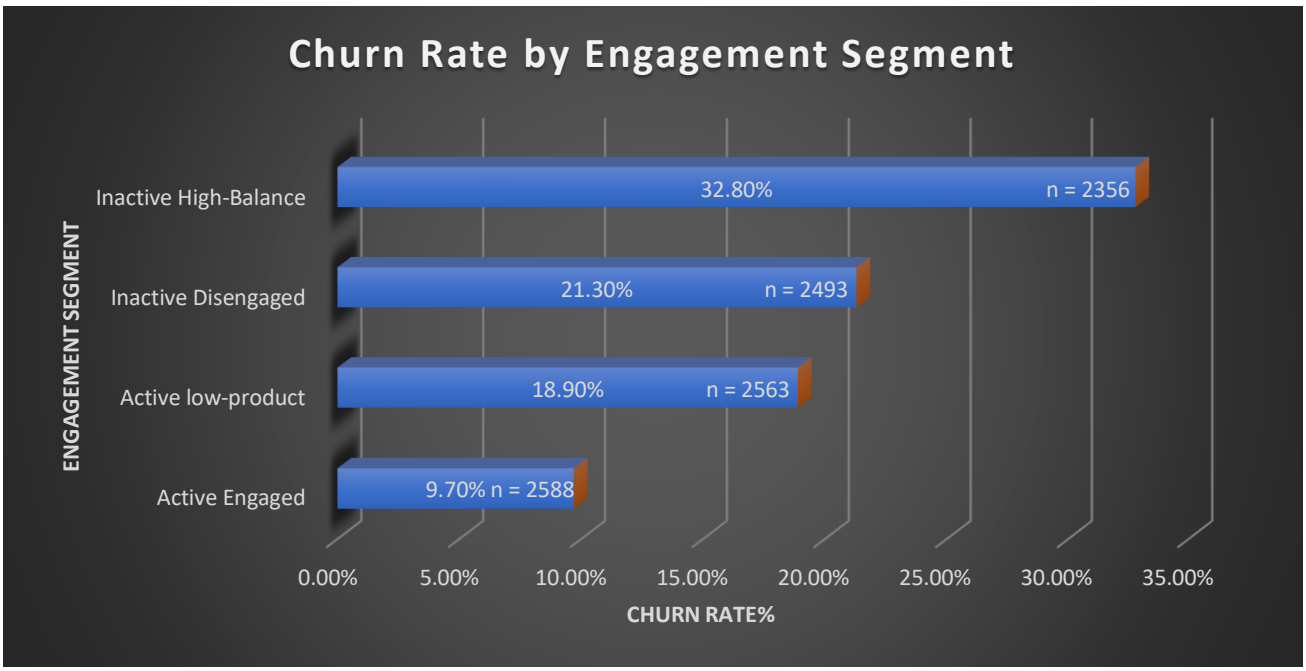
High account balance users have more capital that are being focused by competitors to gain control in a contest to be offered perks, discounts, better rates and performance incentives to ensure loyalty.

PROJECT OBJECTIVES

Evaluate the relationship between engagement and churn

Higher engagement directly drives lower churn rate while reduced engagement serves as the primary indicator of high attrition rate.

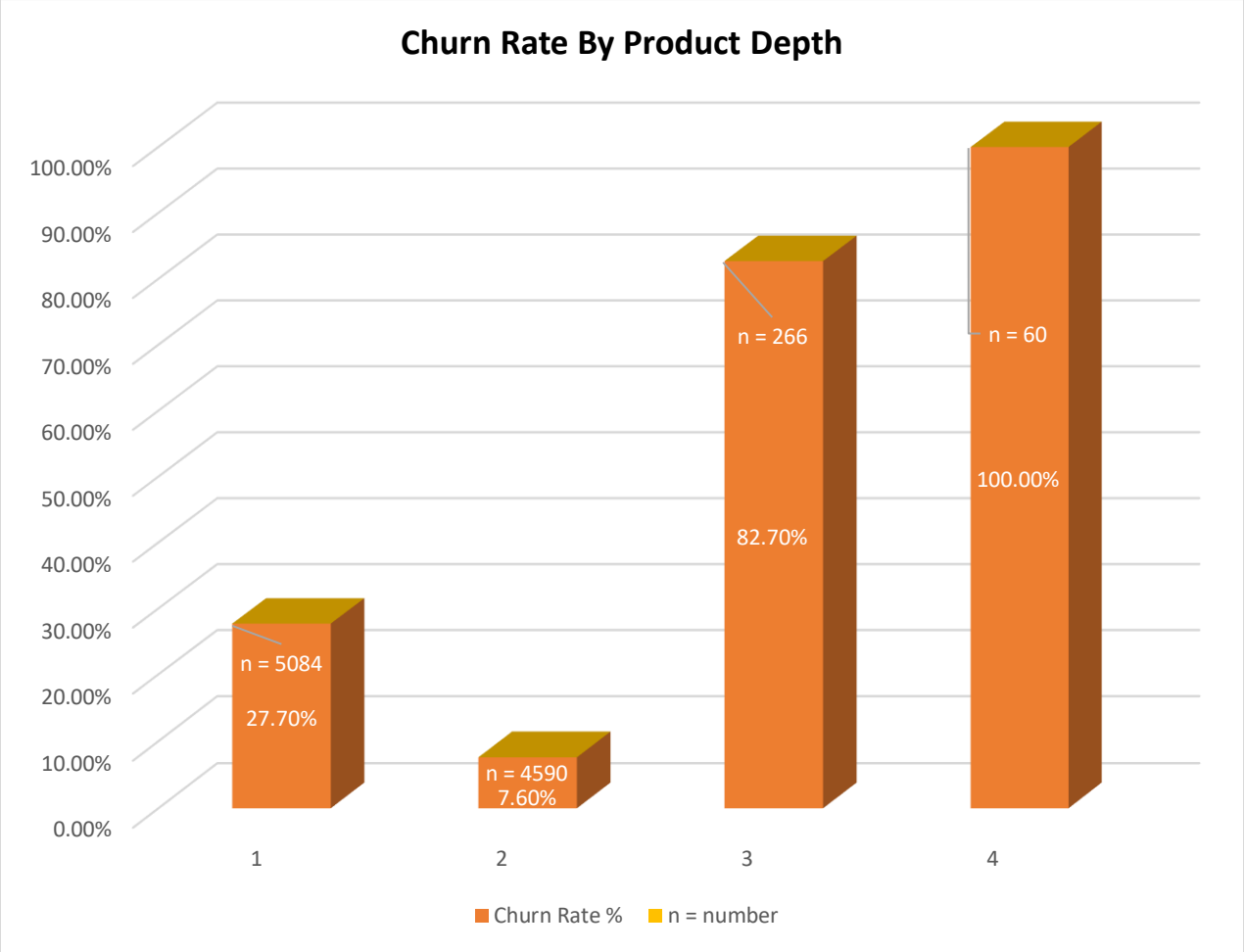
CHURN RATE BY ENGAGEMENT SEGMENT



Measure retention impact of product count and product mix

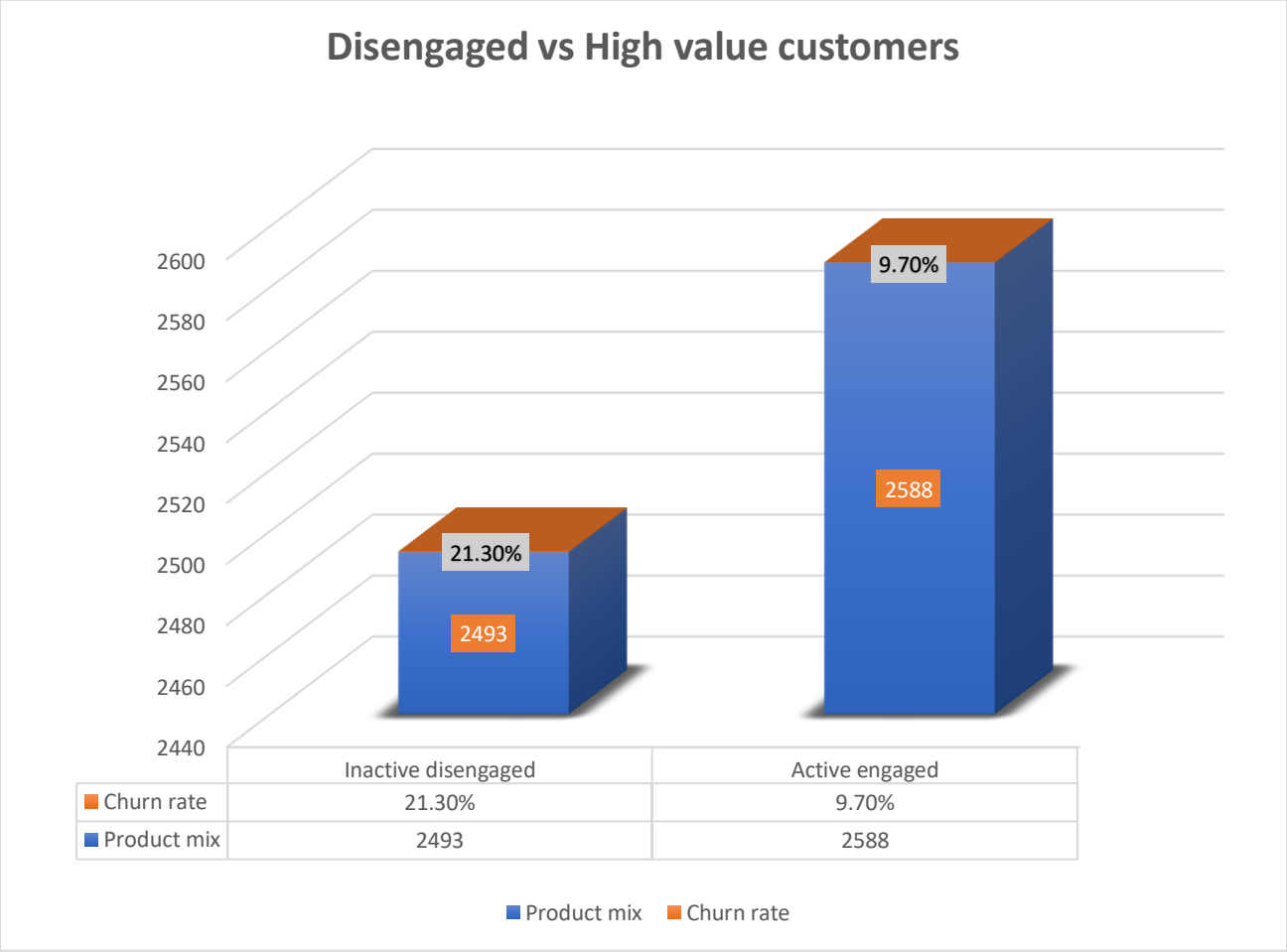
Products Held	Customers	Share	Churn Rate
1	5,084	50.8%	27.7%
2	4,590	45.9%	7.6%
3	266	2.7%	82.7%
4	60	0.6%	100.0%

Impact of product count and product mix



- Customers whose product count is 1 their product mix is 5084 and churn rate is 27.70%.
- Customers whose product count is 2 their product mix is 4590 and churn rate is 7.60%.
- Customers whose product count is 3 their product mix is 266 and churn rate is 82.70%.
- Customers whose product count is 4 their product mix is 60 and churn rate is 100%.

Identify disengaged yet high-value customers



- The value of disengaged customers is 2493 in numbers with an attrition rate of 21.30%.

- Active engaged or high value customers identifies 2588 in numbers with an attrition rate of 9.70%.

SECONDARY OBJECTIVES

Support engagement-driven retention strategies

- Engagement-driven retention strategies support an employee's emotional attachment and motivation directly to focus on company's stability and lower turnover.
- Managers have been provided training as supervisors which accounts upto 70% of the segment in employee retention.
- Daily rewards or recognition based on good performance can increase the likelihood of an employee multiply by 5 times.
- Learning and mentorship programs have been offered to employees as per their demands while applying for internal hiring processes which will promote their career growth.
- Employees fill out the surveys and feedback forms to let us know their concerns and suggestions about the organizational culture.

Improve product bundling decisions

- Product bundling can be either an upsell or a cross sell which will improve decisions based on multiple products that are grouped together to be sold for one combined price.

- A brand might bundle an expensive version of a product with another item to reassure the customers the overall value they would get by buying both the items which is known as upselling.

- A cross sell occurs where the brand bundles related to the products within a certain category. For example, they might sell notebooks with pen which is bundled with each other known as cross selling.

Reduce silent churn among premium customers

Reduce customer churn by finding why customers leave, fixing onboarding, and engaging premium users before they cancel because they forget to cancel their subscription right away.

Arguments will not arise if premium customers will give up due to unreasonable replies.

To reduce silent churn, strategies are taken away to: -

1. **Implement Real-time Behavioural Monitoring**

Monitor customer engagement in real-time that dont wait for monthly analytics reports.

It sets up automated emails when it receives behavioural signals in case of disengagement.

2. **Detect Churn Risk Mid-Interaction**

When a customer mentally gives up during an active session or through a conversation then it is the best time to save a customer to detect churn risk mid interaction.

3. **Monitor friction points in support conversations**

Supported conversations can either be monitored in two ways for or against the frictional moves.

Frustrated languages, repeated issues and negative sentiments against a product that they might stop using it or a service are causes of customer disengagement.

4. **Use session replay to understand user struggles**

An active session between an active user and an end user will save a customer from the struggles that a customer has to face will keep them in a loop to run a replay.

5. Automate proactive outreach

Automated alerts or emails in bulk to endangered customers is to get in touch immediately with helpful resources, personalized support or a check-in call.

6. Create customer health scores

Customer health scores are signals that a customer needs an attention to track customer engagement, product feature adoption, support satisfaction and product usage trends.

DATASET DESCRIPTION

The analysis is based on European_Bank.csv, containing 10,000 customer-level records for a single reference year (2025) across three markets: France, Germany, and Spain.

Each row represents one customer and combines demographic attributes, financial standing, product holdings, and behavioral flags, together with a binary churn indicator (Exited).

No missing values were found in any field, and all binary and categorical fields (Geography, Gender, HasCrCard, IsActiveMember, Exited) were validated for consistent coding prior to analysis.

Field	Description	Type
Customer ID	Unique Customer Identifier	Numeric
Surname	Customer surname	Text
Credit score	Customer creditworthness	Numeric
Geography	France, Germany, Spain	Categorical
Gender	Male / Female	Categorical
Age	Customer Age	Numeric
Tenure	Years with the bank	Numeric
Balance	Account Balance	Numeric (Currency)
NumofProducts	Number of bank products	Numeric
HasCrCard	Credit card ownership	Binary
IsActiveMember	Activity indicator	Binary

The dataset exhibits a class imbalance typical of banking churn data:7,963 customers (79.6%) are retained and 2,037 customers (20.4%) have exited.

Germany exhibits the highest churn rate at 32.4%, followed by France at 16.2% and Spain at 16.7%.

The average customer age is 38.9 years with a standard deviation of 10.5 years.

ANALYTICAL METHODOLOGY (STEP-BY-STEP)

Column	Description	Binary Variables
CustomerId	Unique customer identifier	Numeric
Surname	Customer surname	Text
CreditScore	Customer creditworthiness	Numeric
Geography	France, Spain, Germany	Categorical
Gender	Male / Female	Categorical
Age	Customer age	Numeric
Tenure	Years with the bank	Numeric
Balance	Account balance	Numeric (Currency)
NumOfProducts	Number of bank products	Numeric
HasCrCard	Credit card ownership	Binary
IsActiveMember	Activity indicator	Binary
Estimated salary	Estimated annual salary	Numeric (Currency)
Exited salary	Churn indicator (Target)	Binary

Data ingestion and Data validation

The dataset is loaded using pandas and subjected to a comprehensive validation protocol.

All 14 columns are verified for completeness — the dataset contains zero null values across all variables.

Binary variables (IsActiveMember, HasCrCard, Exited) are confirmed to contain only 0 and 1 values. The churn label

(Exited) is validated as the target variable. Five derived features are engineered during the stage: -

COMMANDS

Step 1: Import Libraries

import streamlit as st import pandas as pd import matplotlib. pyplot as plt import seaborn as sns #

Step 2: Load Dataset
df = pd. read_csv("European_Bank.csv") print (df. shape) # (10000, 14) # —

Step 3: Validation print (df. isnull (). sum ()) # → 0 missing values across all columns
print (df. duplicated (). sum ()) # → 0 duplicate records
print(df['Exited'].value_counts ()) # 7963 Retained|2037 Churned

Step 4: Feature Engineering — RSI
df['RSI'] = (0.4 * df['IsActiveMember'] +

0.3 * (df['NumOfProducts'] / df['NumOfProducts'].max ())

0.3 * (df['Balance'] / df. ['Balance’] .max ())

)

Engagement Classification

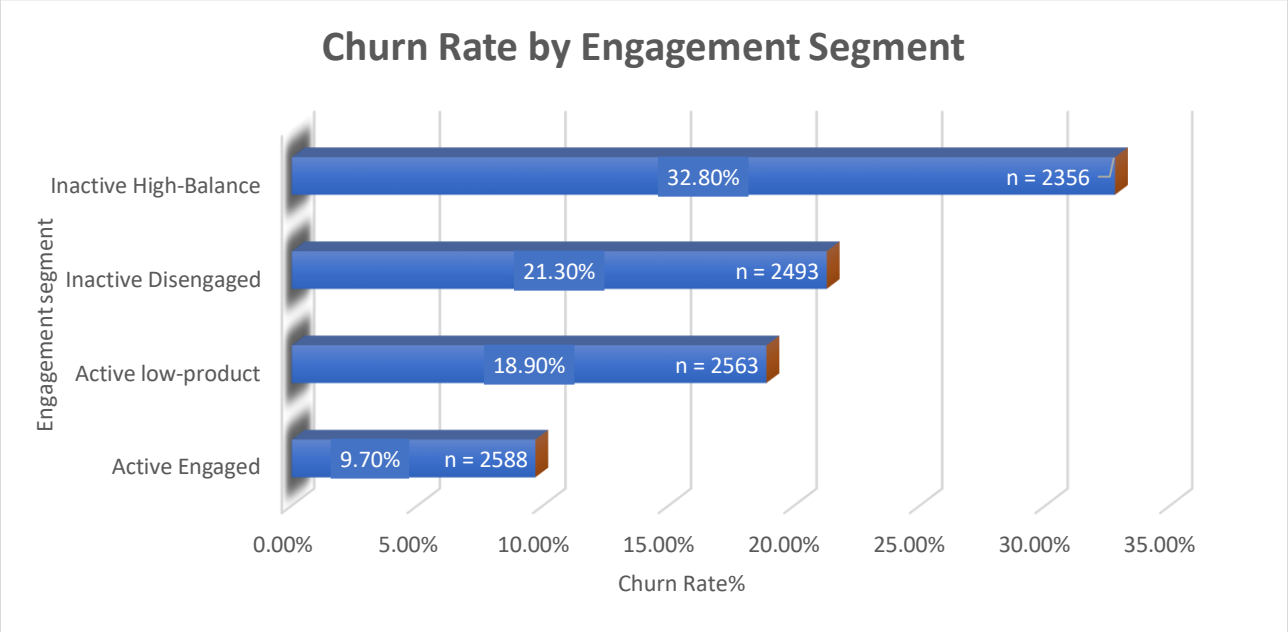


Fig 1.1

Product Utilization Analytics

Product depth — the number of bank products a customer holds — is examined as a primary retention driver.

The product utilization analysis reveals a non-linear relationship between product count and churn, with customers holding 3 or 4 products showing relevated churn rates despite higher product feature adoption.

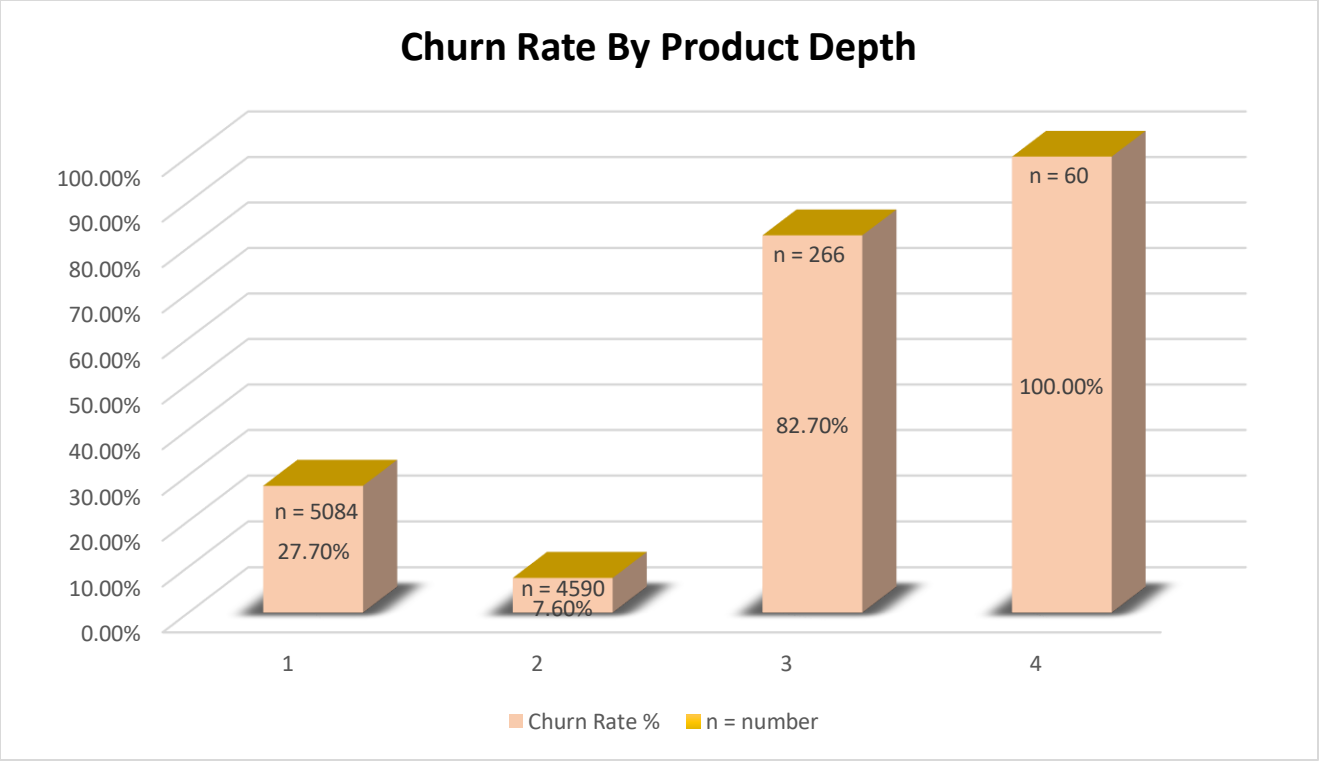


Fig 1.2

Single product vs Multi product

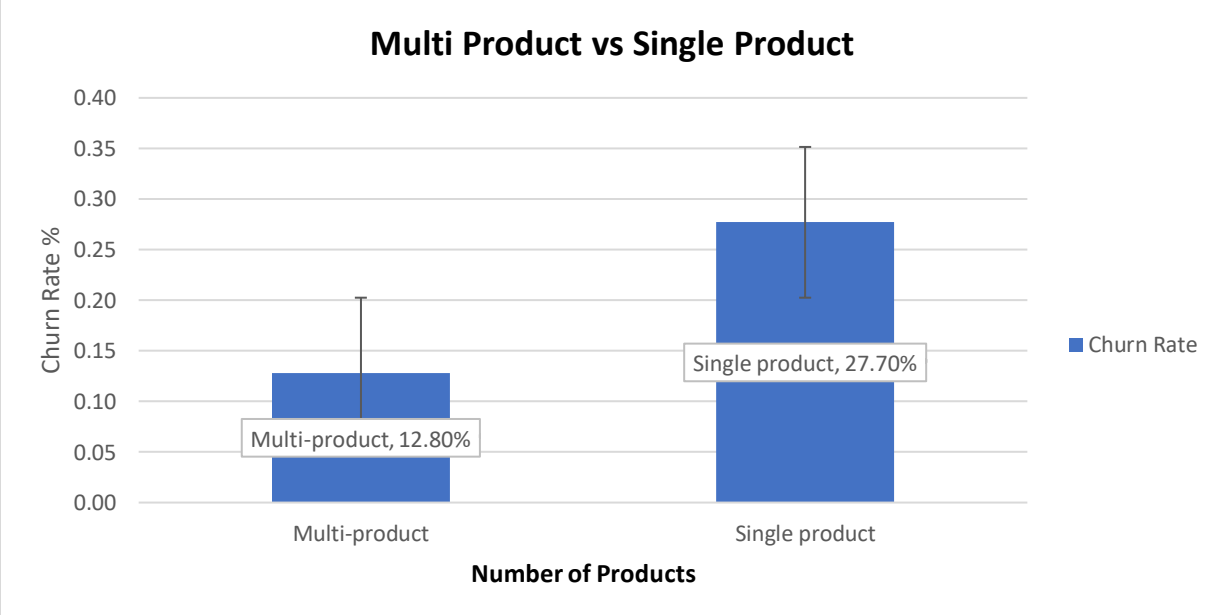


Fig 1.3

FINANCIAL COMMITMENT VS ENGAGEMENT ANALYSIS

Activity status and product depths are engagement analysis ratio that assume loyalty better than financial commitments such as high account balances and salary.

Engagement analysis between inactive disengaged members and active engaged members do have a churn rate 21.30% and 9.70% respectively.

The engagement retention ratio of 21.30% and 9.70% is 2.20%.

The retention ratio is 2.20% which is a financial commitment ratio based on quantitative metrics between inactive and active members.

Balance vs Activity Cross Analysis

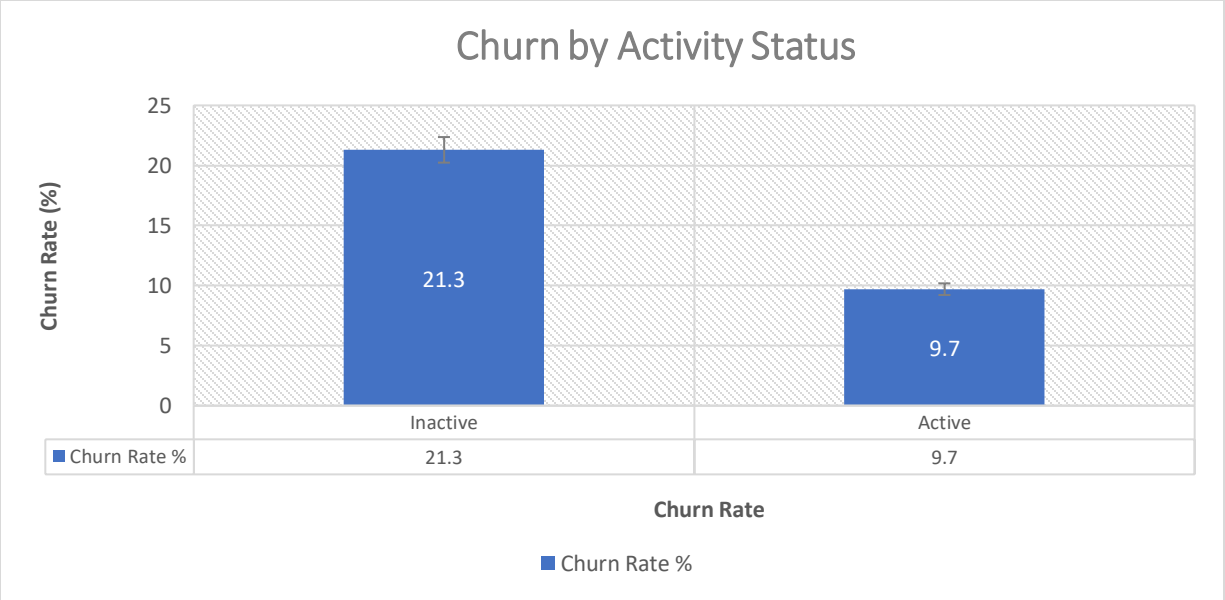


Fig 1.4

Salary-Balance Mismatch Retention

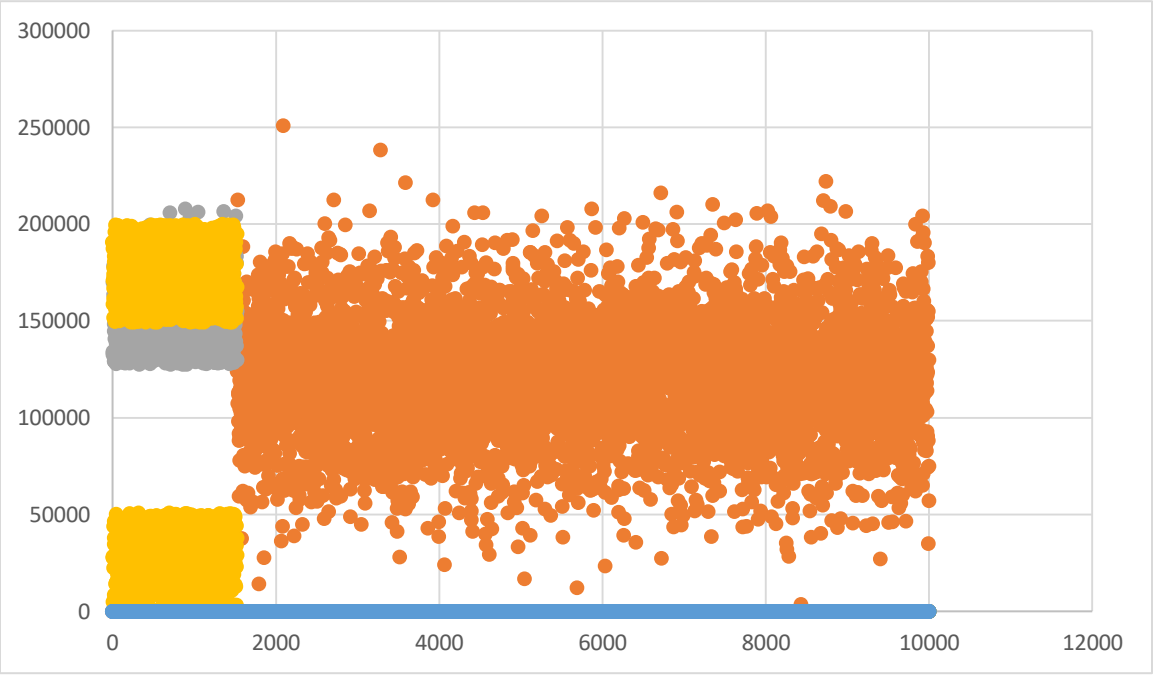


Fig 1.5

Mismatch Rate: - 15.2 % of the current mismatch types.

Definition: - High Balance with Low Salary or Low Balance with High Salary.

Total customers in view = 10000

Churn rate = 20.4 %

Retention rate = 79.6 %

Active customers = 51.5 %

Average credit score = 651

- Sticky customers = 1722 -251 = 1471

Identification of “at-risk premium customers”

From the dataset given, there are few data examples taken for identifying the “at-risk premium customers” group.

At-Risk Customers Analysis

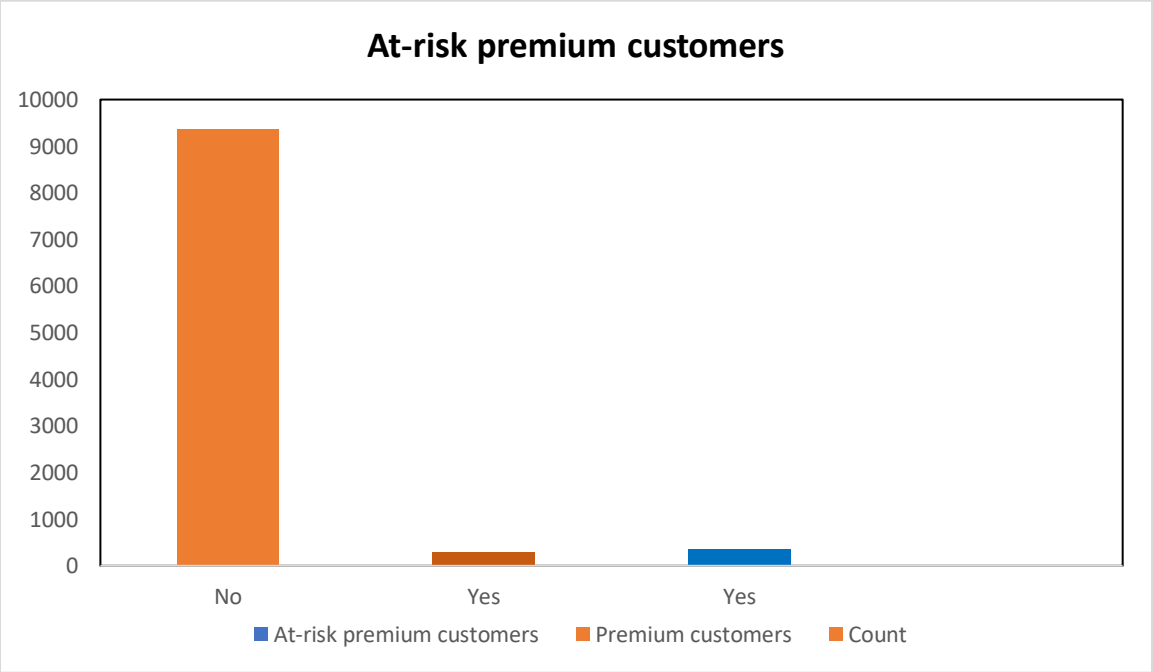


Fig 1.6

At-Risk Premium Customers = 352

Premium Customers = 649

At-Risk Premium = 3.5 %

Premium Customers = 6.5%

Total Customers in view = 10000

RETENTION STRENGTH ASSESSMENT (RSI)

Sticky Customer Analysis

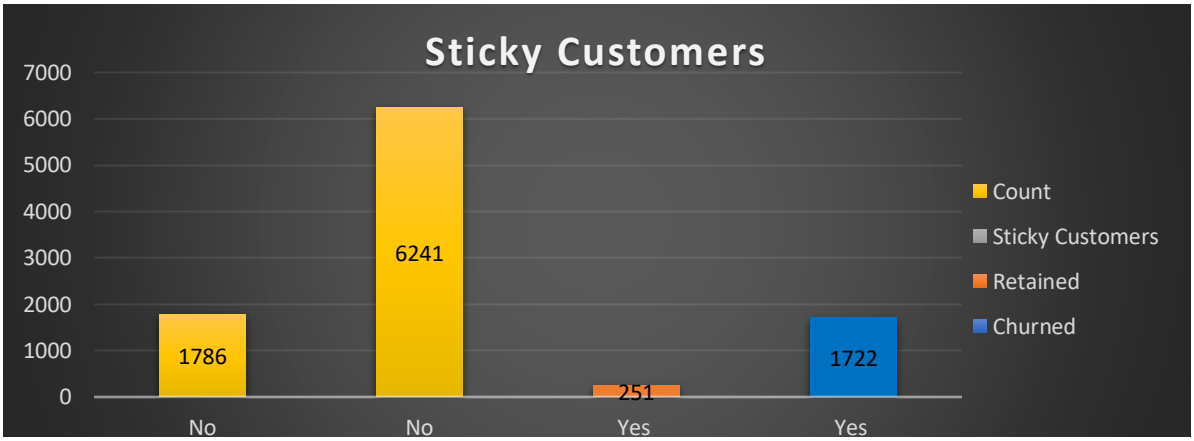


Fig 1.7

Composition of Retention Strength Assessment (RSI)

Headcount of employees (S) = 10000

Retained customers (N) = 251

Headcount of employees (E) = 8027

Subtract retained from the ending headcount: -

8027 – 251 = 7776

Divide by the starting headcount: -

7776 / 10000 = 0.7776 = 0.78

Multiply by 100 to get the percentage :-

0.78 * 100 % = 78%

- The company has an 78% retention rate for the year.

Measure churn stability across Engagement tiers

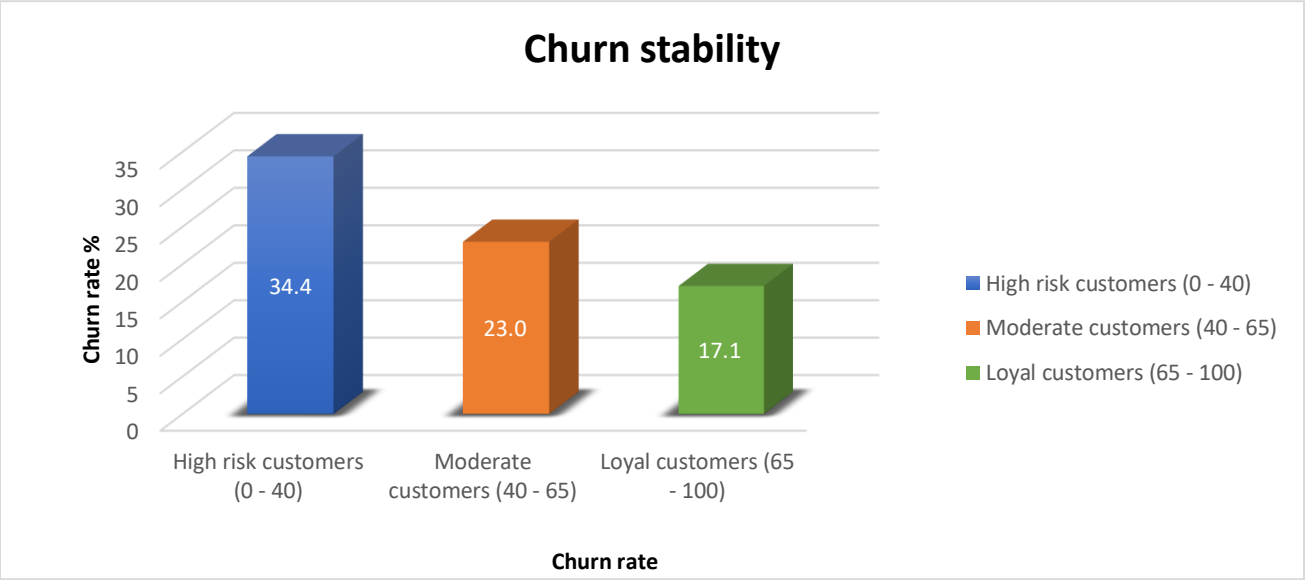


Fig 1.8

High risk customers whose age is between 0 – 40 have a churn stability rate of 34.4%.

Moderate customers whose age is between 40 – 65 have a churn stability rate of 23.0%.

Loyal or sticky customers whose age is between 65 – 100 have a churn stability rate of 17.1%.

Identify Engagement Thresholds linked to Retention

RSI thresholds linked to retention are: -

RSI 0 - 40: High risk customers have an engagement threshold of 34.4% linked to a retention rate of 78%.

RSI 40 – 65: Moderate customers have an engagement threshold of 23.0% linked to a retention rate of 78%.

RSI 65 – 100: Loyal or sticky customers have an engagement threshold of 17.1% linked to a retention rate of 78%.

KEY PERFORMANCE INDICATORS (KPIs)

KPI Name	Description	Values
Engagement Retention Ratio	Active vs Inactive comparison	Active customers = 9.70% Inactive customers = 21.30% Difference = 11.60%
Product depth Index	Products used vs Loyalty	27.70% 7.60% 82.70% 100%
High-Balance Disengagement Rate	Premium churn risk	30.5%
Credit card stickiness score	Card ownership retention impact	Average credit score = 651 Card ownership retention impact = 1722 + 251 = 1973 Stickiness score = 1973/651 = 3.03%
Relationship Strength Index (RSI)	Combined engagement and product score	78%

STREAMLIT WEB APPLICATION REQUIREMENTS

Engagement vs Churn Overview

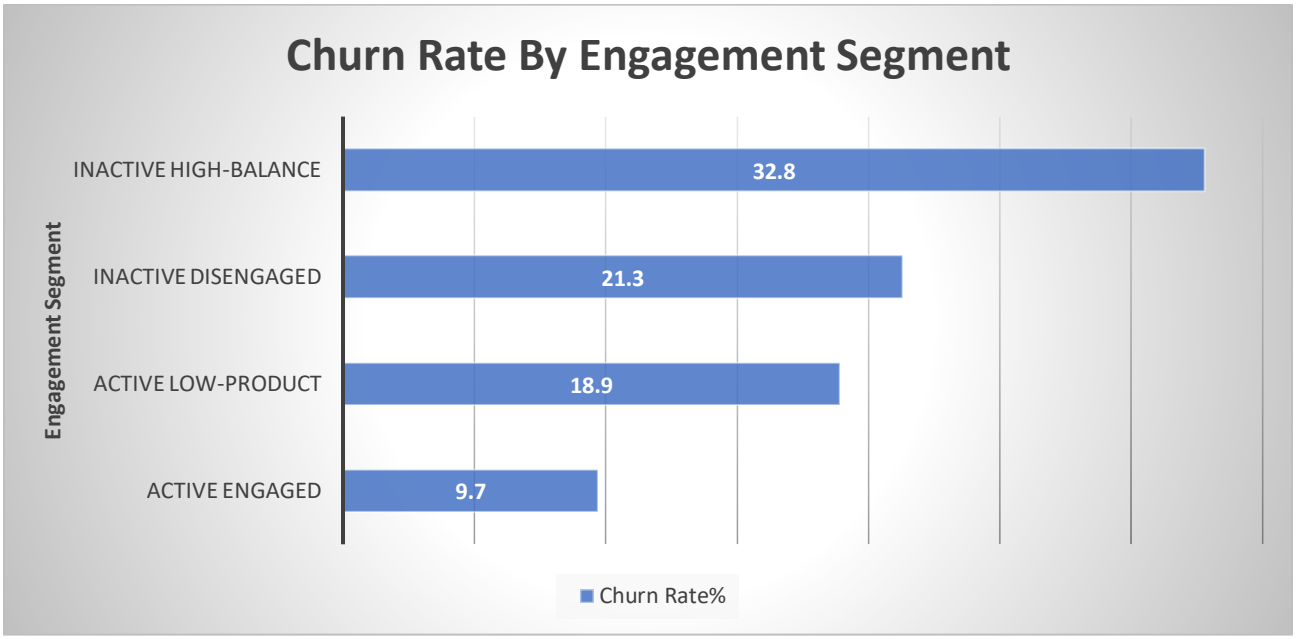
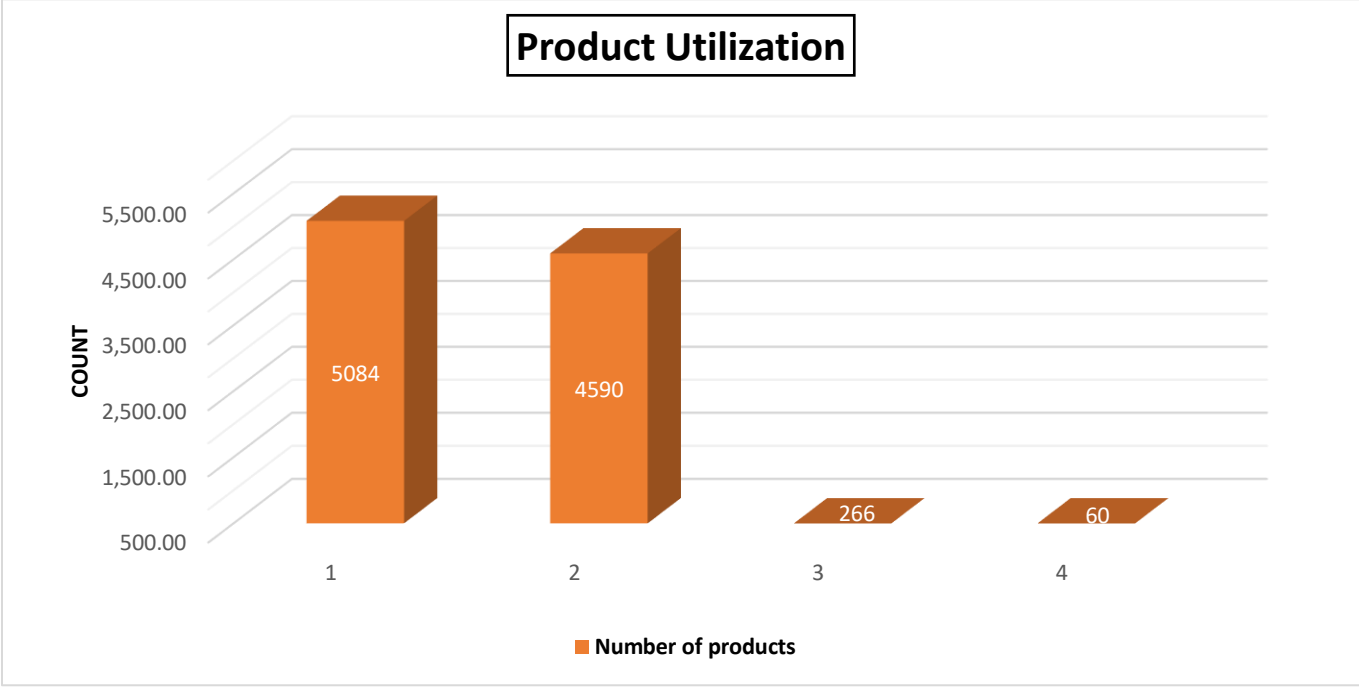


Fig 1.9

Product Utilization Impact Analysis



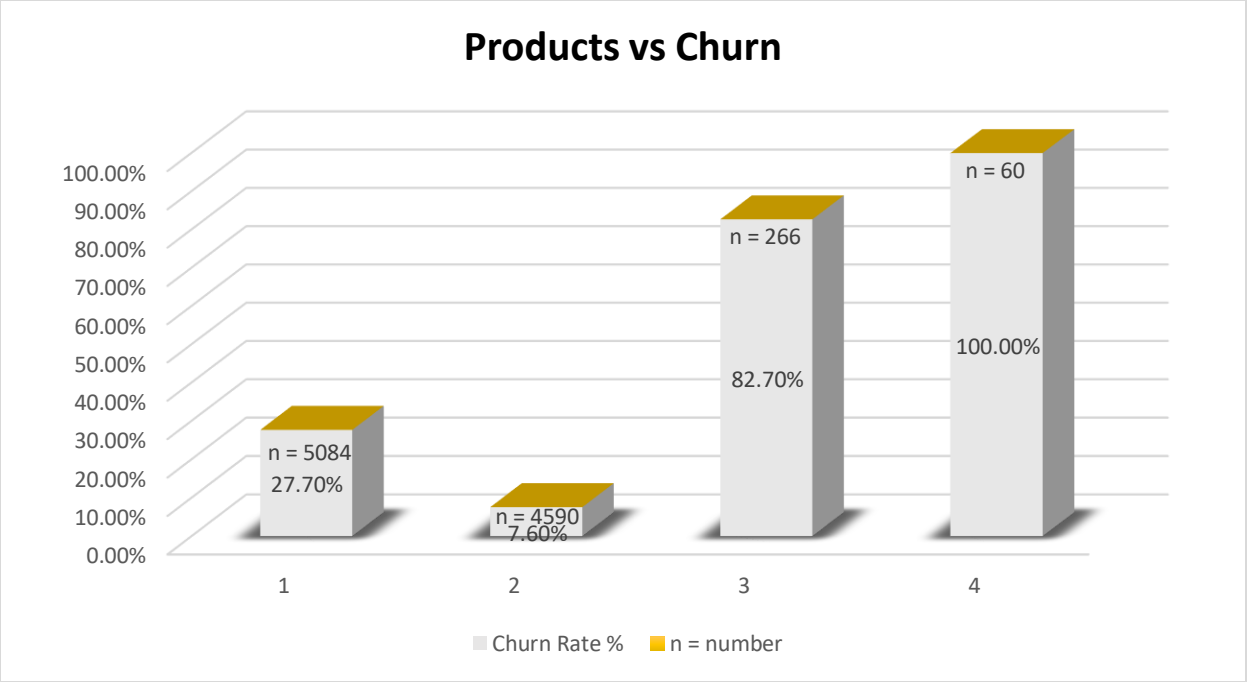

Products vs Churn
100.00%
90.00%
80.00%
70.00%
60.00%
50.00%
40.00%
30.00%
20.00%
10.00%
0.00%
n = 5084
27.70%
n = 4590
7.60%
n = 266
82.70%
n = 60
100.00%
1
2
3
4
Churn Rate %
n = number

Fig 2.0

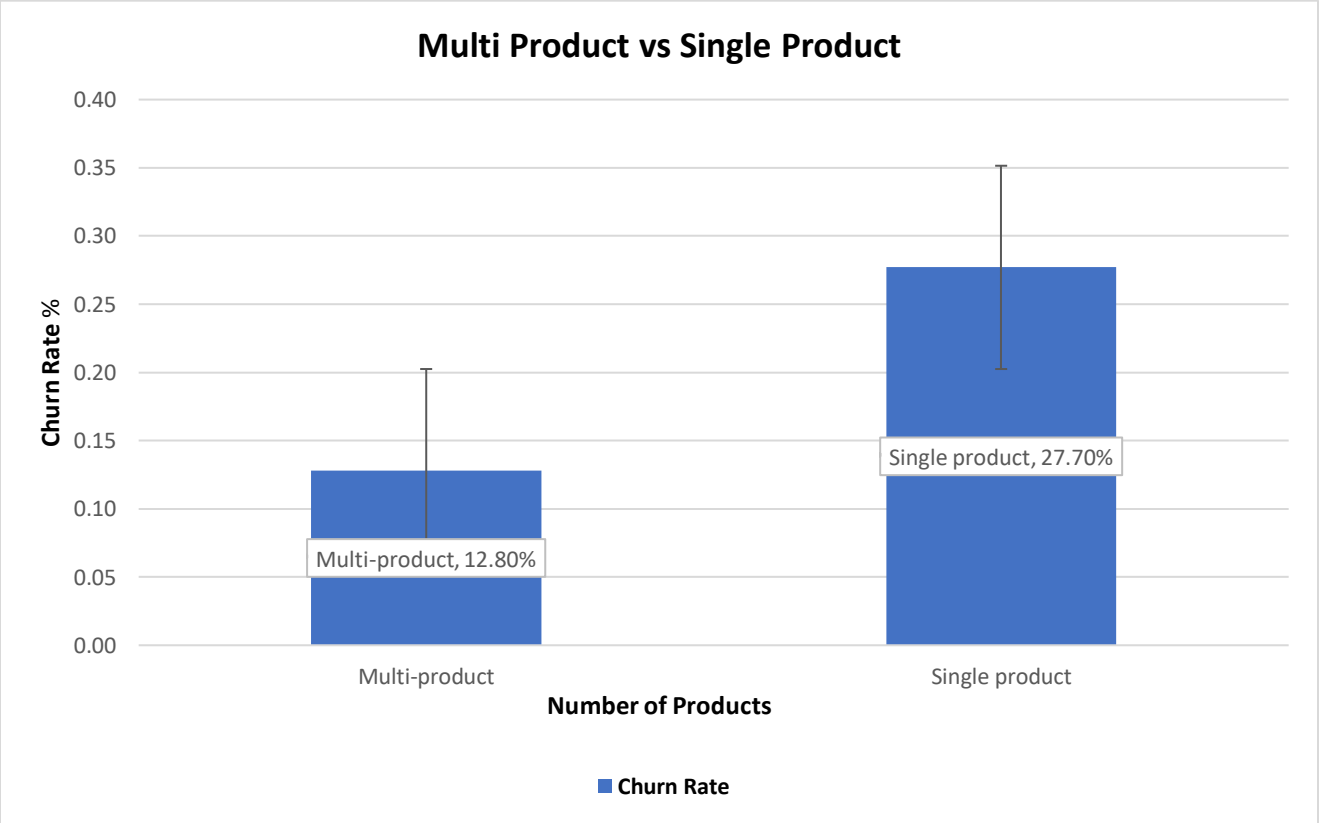

Multi Product vs Single Product
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Churn Rate %
Multi-product, 12.80%
Single product, 27.70%
Multi-product
Single product
Number of Products
Churn Rate

Fig 2.1

High-value Disengaged Customer Detector

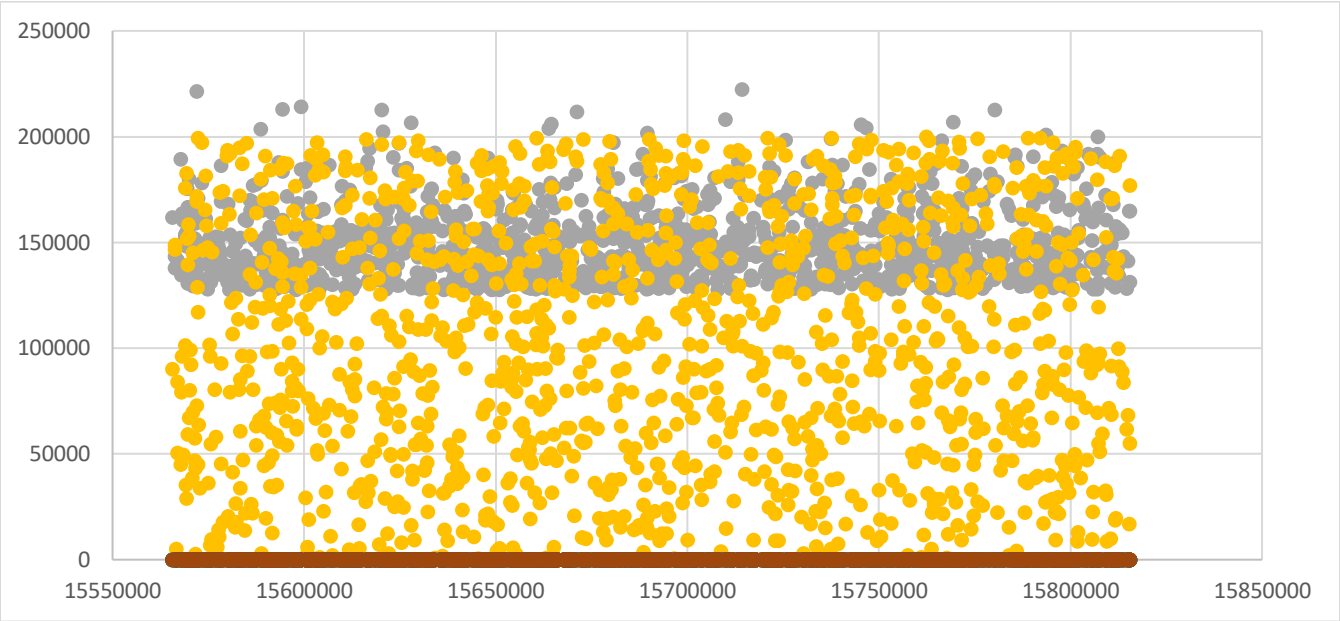
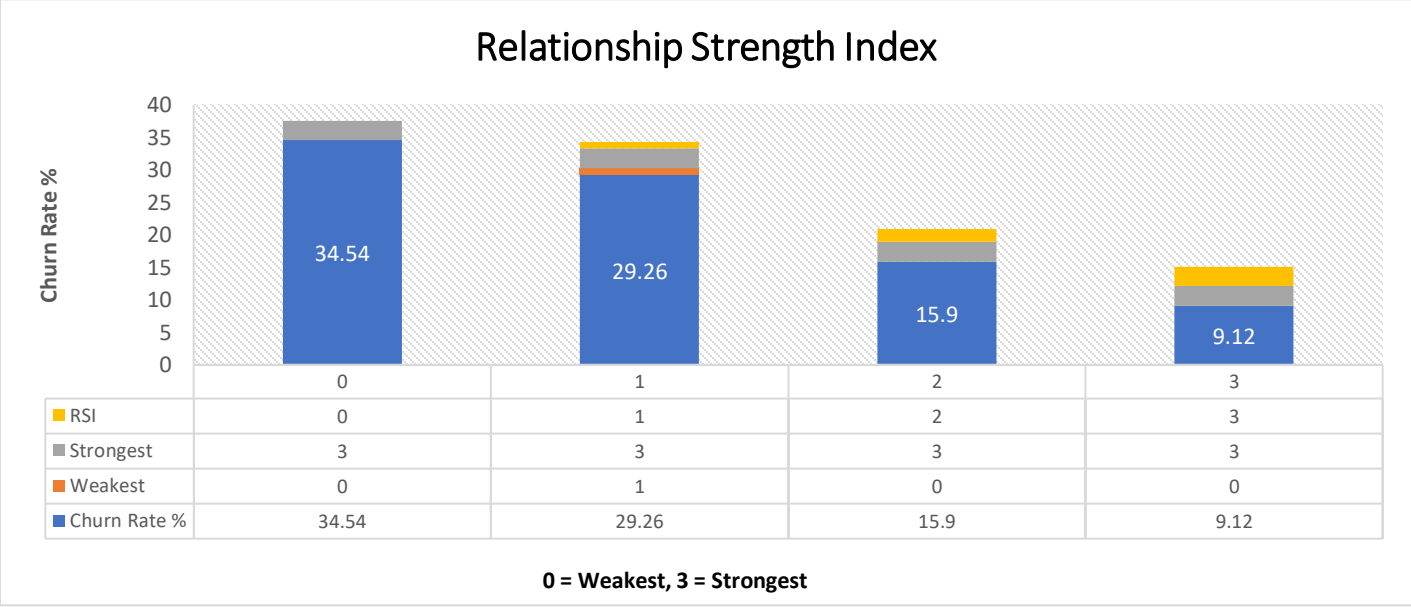


Fig 2.2

- High value Disengaged customers in total = 1248, Churn rate = 30.5%

Retention Strength Scoring Panels



	0	1	2	3
RSI	0	1	2	3
Strongest	3	3	3	3
Weakest	0	1	0	0
Churn Rate %	34.54	29.26	15.9	9.12

Fig 2.3

User Capabilities

Engagement filters

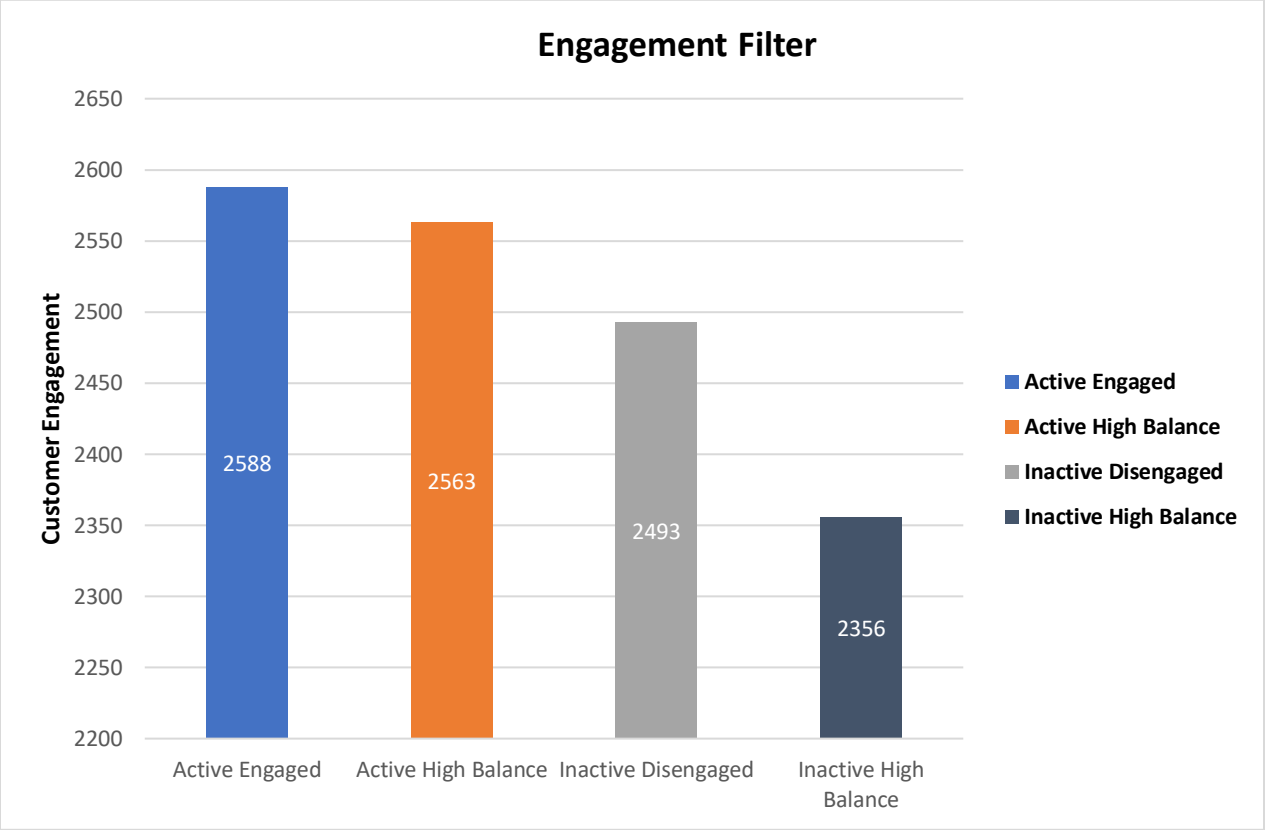


Fig 2.4

From a total of 10000 customers engaged in view, actively engaged customers are 2588 in numbers.

From a total of 10000 customers engaged in view, actively high balance customers are 2563 in numbers.

From a total of 10000 customers engaged in view, inactively disengaged customers are 2493 in numbers.

From a total of 10000 customers engaged in view, inactively high balance customers are 2356 in numbers.

Product Count Sliders

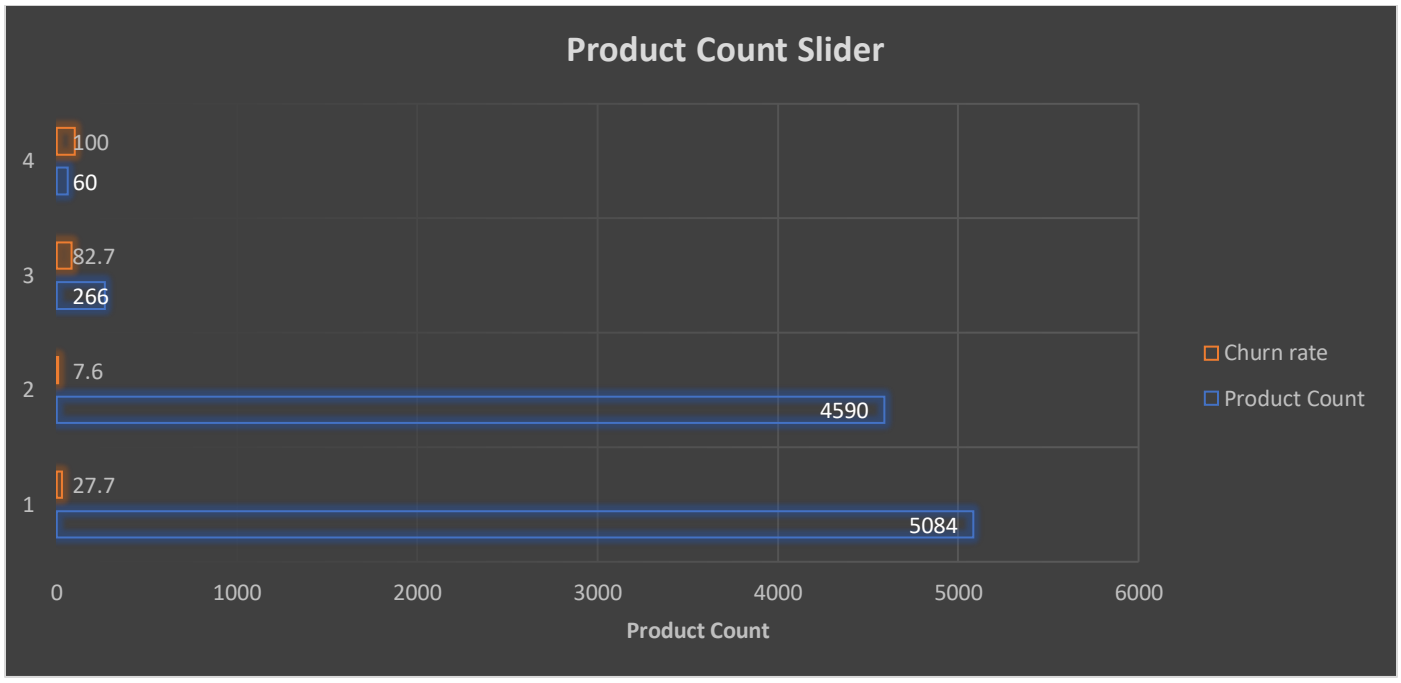


Fig 2.5

Balance and Salary thresholds

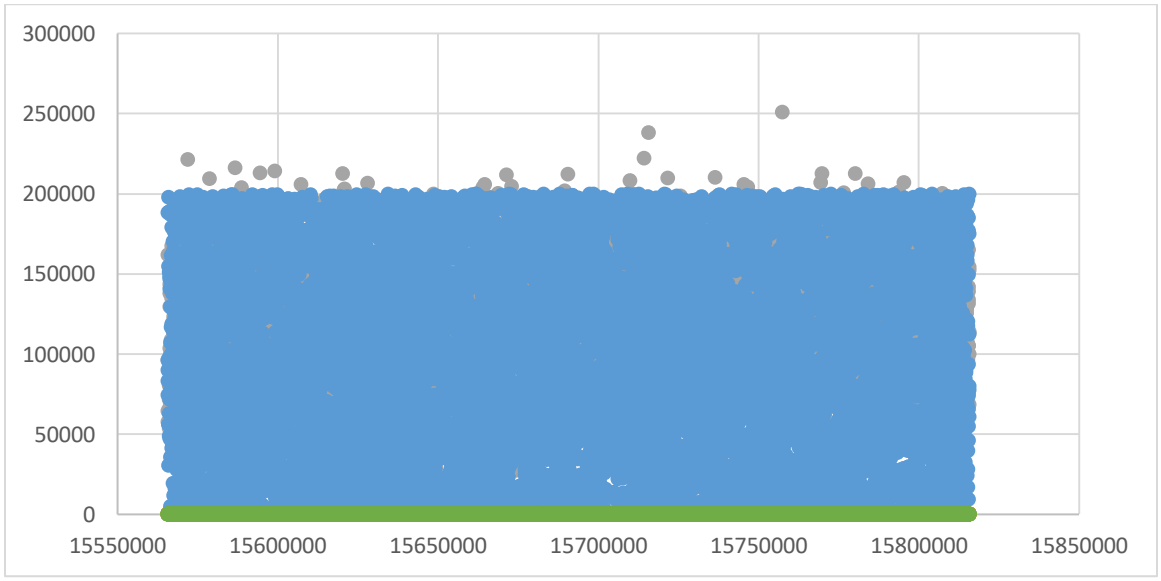


Fig 2.6

This study analyses that with a total of 10000 retail banking accounts, financial thresholds like account balances and

estimated salary puts upto a limit to target the customers either retained or churned.

DELIVERABLES AND SUBMISSION

Research paper (EDA, Insights and Recommendations)

Exploratory data analysis

From Fig1.1, Actively engaged customers are the strongest indicators amongst all churns. Inactive high balance customers are the weakest indicators of amongst all churns.

From Fig 1.2, product depth counts four products into which 60 products have the highest churn rate of 100% and 4590 products has the lowest churn indicator of 7.60%.

From fig 1.3, single product has a churn rate of 27.70% and multi product have a churn rate of 12.80%.

From fig 1.4, Engagement analysis between inactive disengaged members and active engaged members do have a churn rate 21.30% and 9.70% respectively.

The engagement retention ratio of 21.30% and 9.70% is 2.20%.

From fig 1.5, Salary and balance mismatch retention has a mismatch rate of 15.2% which has a churn rate of 20.4% has active members around 51.5% which has an average credit score of 671 with sticky customers around 1471 in numbers.

From fig 2.3, in case of relationship strength scoring panels, the relationship strength index has the strongest churn indicator of 9.12% and has the weakest churn indicator of 34.54%.

CONCLUSION

This study establishes that customer retention in the banking sector is fundamentally driven by behavioural engagement and product utilization rather than financial indicators alone.

The findings demonstrate that customers who actively engage with banking services and utilize multiple products exhibit significantly lower churn rates.

The study demonstrates that customer engagement and product utilization provide stronger indicators of retention than traditional financial measures.

Banks that prioritize engagement-driven strategies and deeper customer relationships can reduce churn, improve retention, and maximize long term customer lifetime value.

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